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PYMBLE LADIES' COLLEGE

YEAR 12

MATHEMATICS EXTENSION I

ASSESSMENT

Time Allowed: 1 hour 30 minutes + 5 minutes reading time

Test date: 7th May, 2001

INSTRUCTIONS:

- All questions should be attempted
- Start each question on a new page
- Approved calculators may be used
- A standard integral sheet is attached
- Write your name and your teacher's name on each page.
- DO NOT staple the questions together
- Hand this question paper in with your answers
- ALL rough working paper must be attached to the back of the last question
- A coloured sheet of paper must be attached, by stapling, to the end of each question
- This assessment task has a value of 15%
- There are 6 questions in this paper
- There are 9 pages in this paper

ASSESSMENT CRITERIA

- **Provide answers which are complete, accurate and comprehensive**
- **Leave your answers in exact form unless otherwise stated**
- **Include all necessary working. Correct answers will not necessarily gain full marks unless necessary working is shown. Relevant working might gain marks even if your answer is wrong.**
- **Take care with mathematical notation**
- **Show relevant information clearly and unambiguously on sketches if required**
- **Present well set out solutions using a logical set of steps in which justification is included where necessary.**

Question 1

- (a) Find the acute angle between the lines $y = -x$ and $\sqrt{3}y = x$ (2 marks)

- (b) Find the indefinite integral of $\int \frac{dx}{\sqrt{1-9x^2}}$ (1 mark)

- (c) If α, β and λ are the roots of the cubic equation $2x^3 + x^2 - x - 2 = 0$ find the value of

(i) $\alpha + \beta + \lambda$

(ii) $\alpha\beta\lambda$

(iii) $\alpha\beta + \alpha\lambda + \beta\lambda$

- (iv) Hence, or otherwise, find the value of $(\alpha - 1)(\beta - 1)(\lambda - 1)$

(4 marks)

- (d) Given $x = 12t$ and $y = 6t^2$ write down

(i) the cartesian equation of the parabola

(ii) the coordinates of the focus

(iii) the equation of the directrix

(3 marks)

Question 2

(Start a new page)

(a) Solve the equation $\sin \theta - \sqrt{3} \cos \theta = 1$ for $0 \leq \theta \leq 2\pi$ (3 marks)

(b) (i) Solve $\sin 2x = \sin x$ for $0 \leq \theta \leq \pi$

(ii) On the same number plane, sketch $y = \sin 2x$ and $y = \sin x$
for $0 \leq \theta \leq \pi$ showing all important features.

(iii) Hence, or otherwise, find the area bounded by the curves
 $y = \sin 2x$ and $y = \sin x$ for $0 \leq \theta \leq \frac{\pi}{3}$

(7 marks)

Question 3

(Start a new page)

(a)

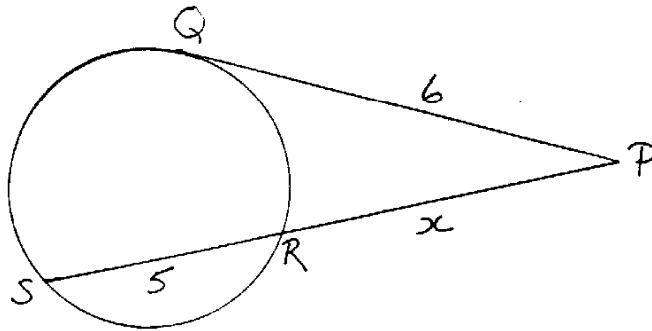


DIAGRAM
NOT TO SCALE

PQ is a tangent to the circle QRS .

PRS is a secant intersecting the circle in R and S

Given that $PQ = 6$, $RS = 5$ and $PR = x$, find x , giving reasons.

(3 marks)

(b)

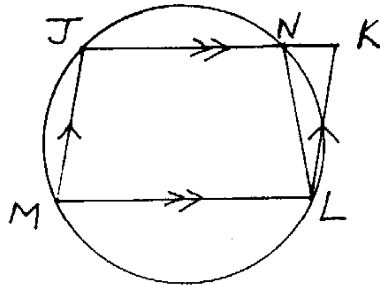


DIAGRAM
NOT TO SCALE

The circle passes through the points J , N , L , and M .

$JKLM$ is a parallelogram

Prove that $NL = LK$, giving reasons.

(3 marks)

(c)

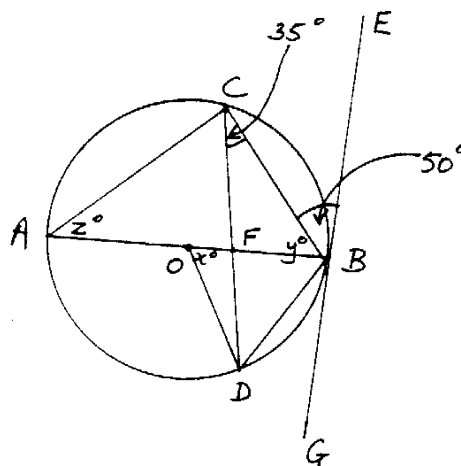


DIAGRAM
NOT TO SCALE

O is the centre of the circle. AB is a diameter.

EBG is a tangent at B . If $\angle CBE = 50^\circ$ and $\angle DCB = 35^\circ$, find the values of x , y and z , giving reasons.

(4 marks)

Question 4

(Start a new page)

(a) Find the general solution for the equation $\tan \theta = \frac{1}{\sqrt{2}}$ (1 mark)

(b) For the function $f(x) = \sqrt{x} + 3$ find

(i) the inverse function

(ii) the domain of the inverse function. (3 marks)

(c) (i) Differentiate $y = \tan^{-1} \frac{1}{x}$, $x \neq 0$

(ii) Hence show that

$$\frac{d}{dx} \left[\tan^{-1} x + \tan^{-1} \frac{1}{x} \right] = 0$$

(iii) Then sketch the curve $y = \tan^{-1} x + \tan^{-1} \frac{1}{x}$ (6 marks)

Question 5

(Start a new page)

- (a) The function $f(x) = x^3 - x^2 - x - 1$ has a zero between 1 and 2
- (i) Taking $x = 2$ as a first approximation to this zero, use Newton's method to calculate a second approximation.
- (ii) Would $x = 1$ have been a suitable first approximation to use? Explain your answer fully.

(4 marks)

- (b) The function $f(x) = ax^3 + bx^2 + cx + d$ has a double zero at $x = 1$ and a minimum value of -4 when $x = -1$.

Find the values of a , b , c and d .

(6 marks)

Question 6

(Start a new page)

Consider the parabola $x^2 = 4ay$

- (i) Find the equation of the tangent at $P(2ap, ap^2)$
- (ii) If the tangent at P cuts the y -axis at T , show that $T = (0, -ap^2)$
- (iii) Find the equation of the normal at P .
- (iv) If the normal at P cuts the y -axis at N , show that $N = (0, 2a + ap^2)$
- (v) Explain why NT is the diameter of the circle passing through PTN . Hence find the equation of the circle.
- (vi) If the tangent at P cuts the x -axis at R and M is the midpoint of RN , show that $R = (ap, 0)$ and find the co-ordinates of M .
- (vii) Determine the equation of the locus of M and describe this locus geometrically.

(10 marks)

END OF PAPER

Solution dy

+1 Marks
Answer

Qn-1

(a) $y = -x$
 $\therefore m = -1$

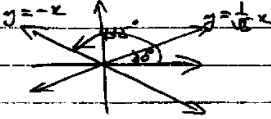
$\sqrt{3}y = x$
 $y = \frac{x}{\sqrt{3}}$

$\therefore m = \frac{1}{\sqrt{3}}$

$\therefore \alpha = 135^\circ$

$\therefore \beta = 30^\circ$

$\therefore \text{acute angle} = 180^\circ - (135^\circ - 30^\circ)$
 $= 180^\circ - 105^\circ$
 $= 75^\circ$



(b) $\int \frac{dx}{1+9x^2}$

$= \frac{1}{3} \sin^{-1}(3x) + C$

(c) $2x^3 + x^2 - x - 2 = 0$

(i) $\alpha + \beta + \lambda = -\frac{1}{2}$

(ii) $\alpha\beta\lambda = \frac{2}{2} = 1$

(iii) $\alpha\beta + \alpha\lambda + \beta\lambda = -\frac{1}{2}$

(iv) $(\alpha-1)(\beta-1)(\lambda-1)$

$= (\alpha\beta - \alpha - \beta + 1)(\lambda - 1)$

$= \alpha\beta\lambda - \alpha\lambda - \beta\lambda + \lambda - \alpha\beta + \alpha + \beta - 1$

$= \alpha\beta\lambda - (\alpha\lambda + \beta\lambda + \alpha\beta) + (\alpha + \beta + \lambda) - 1$

$= 1 + \frac{1}{2} + (-\frac{1}{2}) - 1$

$= 0$

Qn-1

(i) $x = 12t \Rightarrow t = \frac{x}{12}$
 $y = 6t^2$

$\therefore y = 6 \cdot \frac{x^2}{144}$
 $\therefore x^2 = 24y$

(ii) $x^2 = 24y$

$\therefore x^2 = 4(6)y$

$\therefore \text{Focus is } (0, 6)$

(iii) Eqn. of directrix is $y = -6$

Q11.2

6.1 $\sin \theta - \sqrt{3} \cos \theta = 1$, $0 \leq \theta \leq 2\pi$

$2 \sin(\theta - \frac{\pi}{3}) = 1$

$\sin(\theta - \frac{\pi}{3}) = \frac{1}{2}$

$\theta - \frac{\pi}{3} = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$

$\therefore \theta = \frac{\pi}{2} \text{ or } \frac{7\pi}{6}$

$r = \sqrt{1^2 + (\sqrt{3})^2}$

$r = 2$

$\tan \theta = \sqrt{3}$

$\theta = \frac{\pi}{3}$

$0 \leq \theta \leq 2\pi$

$-\frac{\pi}{3} \leq \theta - \frac{\pi}{3} \leq 2\pi - \frac{\pi}{3}$

(b) (i) $\sin 2x = \sin x$, $0 \leq x \leq \pi$

$\sin 2x - \sin x = 0$

$2 \sin x \cos x - \sin x = 0$

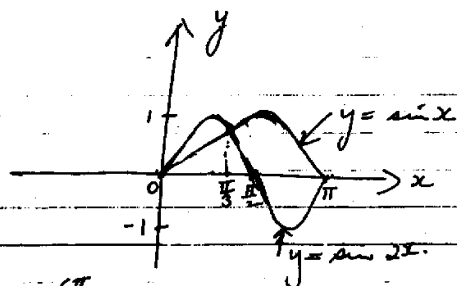
$\sin x (2 \cos x - 1) = 0$

$\therefore \sin x = 0 \text{ or } \cos x = \frac{1}{2}$

$\therefore x = 0 \text{ or } \pi \text{ or } x = \frac{\pi}{3}$

$\therefore x = 0, \frac{\pi}{3} \text{ or } \pi$

(ii)



(iii) Area = $\int_0^{\pi/3} (\sin 2x - \sin x) dx$

= $\left[-\frac{\cos 2x}{2} + \cos x \right]_0^{\pi/3}$

= $\left(-\frac{1}{2} + \frac{1}{2} \right) - \left(-\frac{1}{2} + 1 \right) = \frac{1}{4} \text{ unit}^2$

Q3 a) $6^2 = x(5+x)$ (square of tangent = product of intercepts)

$x^2 + 5x - 36 = 0$

$x = 4 \text{ or } -9$

but $x > 0$ (x is a length)

$\therefore x = 4$

b) $\hat{KNL} = \hat{JML}$ (ext. $\angle$ = int. opp in cyclic quad)

$\hat{NKL} = \hat{JML}$ (opp $\angle$'s of parallelogram)

$\therefore \triangle NKL$ is isosceles ($\hat{NKL} = \hat{KNL}$ base angles equal)

$\therefore NL = KL$ (sides opp equal angles of $\triangle$)

c) $z = \hat{CBE}$ ($\angle$ between chord and tangent = $\angle$ in alternate segment)

= 50°

$\hat{ABE} = 90^\circ$ (tangent $\perp$ to radius)

$\therefore y + 50^\circ = 90^\circ$

$y = 40^\circ$

$x = 2 \times \hat{FCB}$ ($\angle$ at centre = twice $\angle$ at circumference)

= $2 \times 35^\circ$

= 70°

Common errors: incomplete or missing reasons
false statements, especially in part (a).

Qu-4

(a) $\tan \theta = \frac{1}{\sqrt{2}}$

$\therefore \theta = n\pi + \tan^{-1} \frac{1}{\sqrt{2}}$

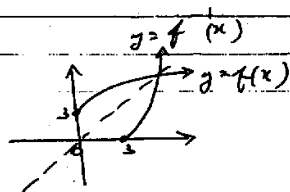
(b) $f(x) = \sqrt{x} + 3$

(i) $y = \sqrt{x} + 3$

$y - 3 = \sqrt{x}$

$x = (y - 3)^2$

$\therefore f^{-1}(x) = (x - 3)^2, x \geq 3, y \geq 0$



(ii) Domain: $x \geq 3$

(c) (i) $y = \tan^{-1} \frac{1}{x}, x \neq 0$

$y' = \frac{-x^{-2}}{1 + (\frac{1}{x})^2}$

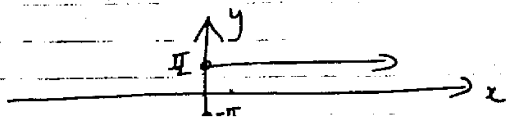
$= \frac{-\frac{1}{x^2}}{\frac{x^2 + 1}{x^2}}$

$= \frac{-1}{x^2 + 1}$

(ii) $\frac{d}{dx} \left[\tan^{-1} x + \tan^{-1} \frac{1}{x} \right]$

$= \frac{1}{1+x^2} + \left(\frac{-1}{1+x^2} \right)$

$= 0$



(iii)

Qu-5

(a) (i) $f(x)$ has a zero between 1 & 2.

$f'(x) = 3x^2 - 2x - 1$

then $x = 2 - \frac{1}{3(4) - 2(2) - 1}$

$= 2 - \frac{1}{7}$

$= 1\frac{6}{7}$

(ii) take $x = 1$ as 1st approxⁿ

then $x = 1 - \frac{-2}{3(1)^2 - 2(1) - 1}$

$= 1 - \frac{2}{3 - 2 - 1}$

$= 1 - \frac{2}{0}$

Since $f'(1) = 0$, $f(x)$ has a stat. pt. at $x = 1$. Hence it is unsuitable as a first approxⁿ.

* and therefore tangent at $x = 1$ is parallel to x axis

Q4-5.

(i) $f(x) = ax^3 + bx^2 + cx + d$
 $f(1) = a + b + c + d = 0 \quad \dots (1)$
 $f(-1) = -a + b - c + d = -4 \quad \dots (2)$
 $f'(x) = 3ax^2 + 2bx + c$
 $f'(-1) = 3a - 2b + c = 0 \quad \dots (3)$
 $f'(1) = 3a + 2b + c = 0 \quad \dots (4)$

(3) - (4) $\Rightarrow -4b = 0$
 $\therefore b = 0$

Subst for b in (1), (2) & (3)

(1) $\Rightarrow a + c + d = 0 \quad \dots (5)$
(2) $\Rightarrow -a - c + d = -4 \quad \dots (6)$
(3) $\Rightarrow 3a + c = 0 \quad \dots (7)$

(5) + (6) $\Rightarrow 2d = -4$
 $d = -2$

Subst. for d in (5)

$a + c - 2 = 0$

$a + c = 2$

Subst (7) $a = 2 - c \quad \dots (8)$

$3(2 - c) + c = 0$

$6 - 3c + c = 0$

$6 - 2c = 0$

$2c = 6$

$c = 3$

$a + c - 2 = 0$

$a + 3 - 2 = 0$

$\therefore a = -1, b = 0$

$\therefore c = 3, d = -2$

Q4-6

(i) $x^2 = 4ay$
 $y = \frac{1}{4a} x^2$
 $y' = \frac{1}{2a} x$

When $x = 2ap$, $y' = p$

Eqn of tangent at P is:

$p = \frac{y - ap^2}{x - 2ap}$

$px - 2ap^2 = y - ap^2$

$\therefore y = px - ap^2$

(ii) When $x = 0$
 $y = 0 - ap^2 = -ap^2$
 $\therefore T = (0, -ap^2)$

(iii) $m(\text{normal}) = -\frac{1}{p}$
Eqn of normal at P is:
 $-\frac{1}{p} = \frac{y - ap^2}{x - 2ap}$

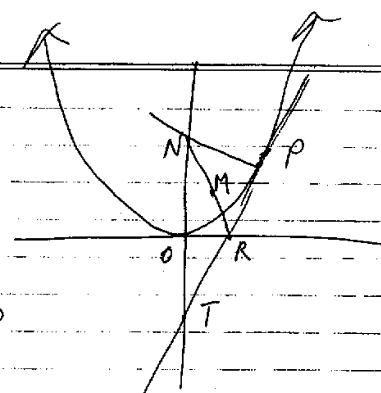
$-x + 2ap = py - ap^3$

$x + py - 2ap - ap^3 = 0$

When $x = 0$
 $py = 2ap + ap^3$

$y = 2a + ap^2$

$\therefore N = (0, 2a + ap^2)$



Q. 6 (contd.)

• (v) $\angle NPT = 90^\circ$ (angle between tangent + normal is 90°)

• 4 an angle in a semicircle is 90°
 $\therefore NT$ is a diameter of the circle passing thro P

$$\begin{aligned}\text{Centre of this circle} &= \left(0, \frac{2a + ap^2 - ap^2}{2}\right) \\ &= (0, a)\end{aligned}$$

$$\text{Radius} = \frac{2a + ap^2 + ap^2}{2} = a + ap^2$$

• Eqn of circle is:

$$x^2 + (y - a)^2 = (a + ap^2)^2$$

• (vi) Eqn. of tangent at P is: $y = px - ap^2$

$$\begin{aligned}\text{when } y = 0, \quad px - ap^2 &= 0 \\ px &= ap^2 \\ x &= ap\end{aligned}$$

$$\therefore R = (ap, 0)$$

$$\begin{aligned}\therefore M &= \left(\frac{0 + ap}{2}, \frac{2a + ap^2 + 0}{2}\right) \\ &= \left(\frac{ap}{2}, \frac{2a + ap^2}{2}\right)\end{aligned}$$

$$\text{(vii)} \quad \begin{cases} x = \frac{1}{2}ap \\ y = a + \frac{1}{2}ap \end{cases}$$

Subst. $p = \frac{2x}{a}$ into y

$$\begin{aligned}y &= a + \frac{1}{2}a \left(\frac{2x}{a}\right) \\ &= a + \frac{1}{2}a \left(\frac{4x^2}{a^2}\right) \\ &= a + \frac{2x^2}{a}\end{aligned}$$

$$2x^2 = a(y - a)$$

$$x^2 = \frac{a}{2}(y - a)$$

$\therefore$ locus of M is a parabola with vertex at $(0, a)$ + a focal length at $\frac{1}{2}a$